Jump App User Guide

CSC1049 - Year 3 Project
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1. What is Jump?

Jump is a web app we built to show how reliable Irish public transport actually is. Most apps just show you the timetable, but buses and trains don't always stick to it. Jump pulls in live data from the NTA (National Transport Authority) and works out how often services actually run on time.

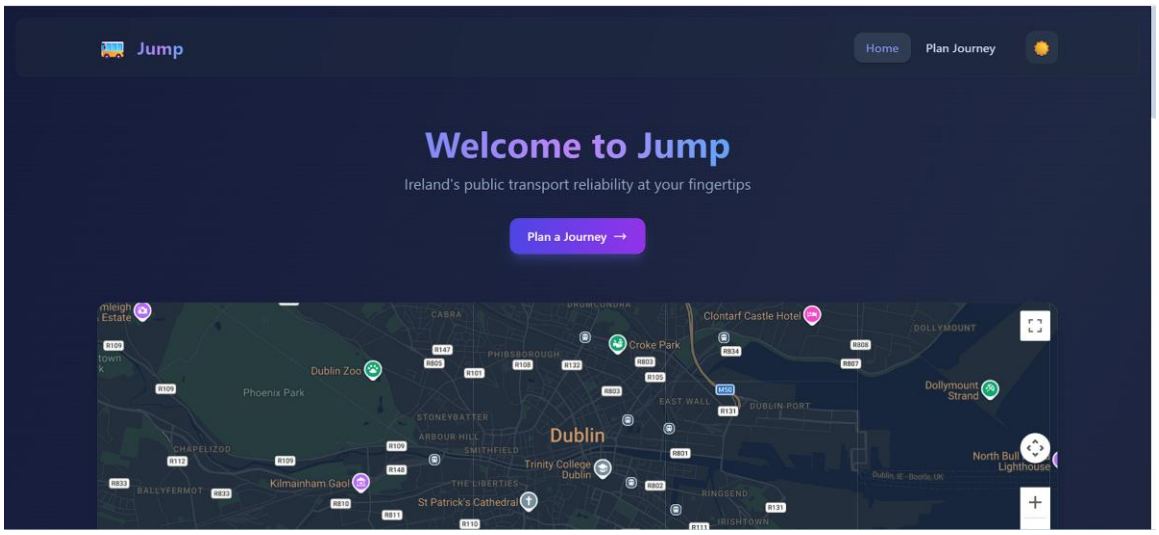
You can look up any route or stop in the country, plan a journey and see how reliable each option is, save routes you use a lot, and filter reliability by day and time. It covers Dublin Bus, Bus Eireann, Irish Rail, Luas, Go-Ahead, Local Link, and loads of smaller regional operators too.

2. Opening the App

Open any browser and go to http://localhost:3000. There's no login or signup needed, it just loads straight in.

At the top of every page there's a nav bar with the Jump logo (click it to get back to the home page), a Home link, a Plan Journey link, and a little moon/sun icon for switching between light and dark mode.

If you get a "Connection Error" when you open it, the backend server probably isn't running. Check the Common Issues section at the end.



3. Home Page

This is the main landing page. There's a big "Plan a Journey" button at the top if you want to jump straight into route planning.

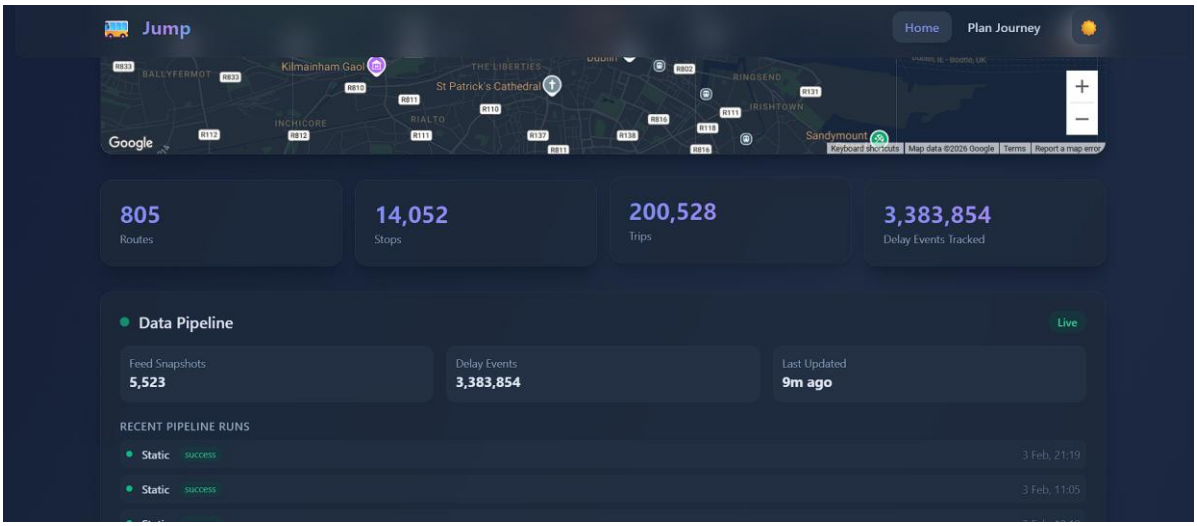
If you've saved any favourite routes, they show up right below that. Each one is a clickable card that brings you to the route details. You can remove them by hitting the heart icon.

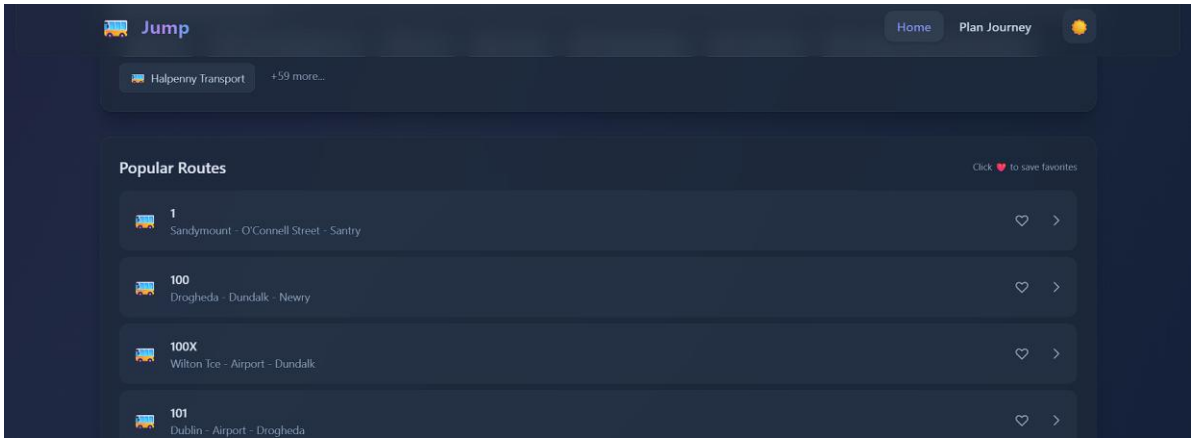
There's a map of Ireland underneath, and then four stat cards showing how many routes, stops, trips, and delay events we're tracking.

The Data Pipeline box tells you if we're actively pulling in live data or not. If it says "Live" with a green dot, data is being collected. It also shows things like how many snapshots have been taken, total delay events, when the last update was, and a list of recent pipeline runs.

Below that there's a breakdown of transport types (bus, rail, tram) and then a list of all the operators. The big ones like Dublin Bus and Irish Rail are shown first, then Local Link services, and then you can expand to see all the smaller operators.

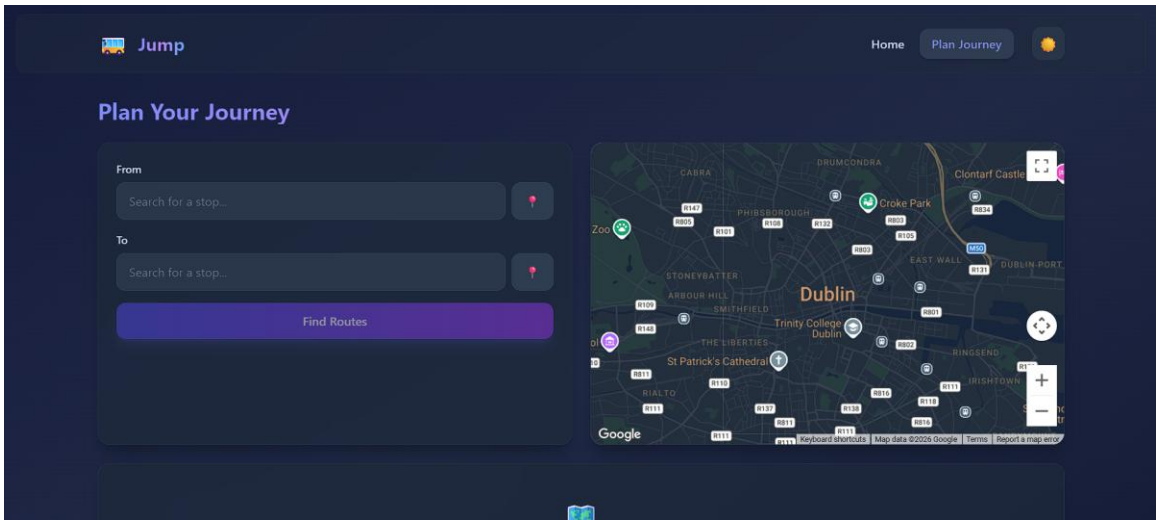
At the bottom there's a list of routes. You can click the heart to favourite them or click the route itself to see its details.





4. Planning a Journey

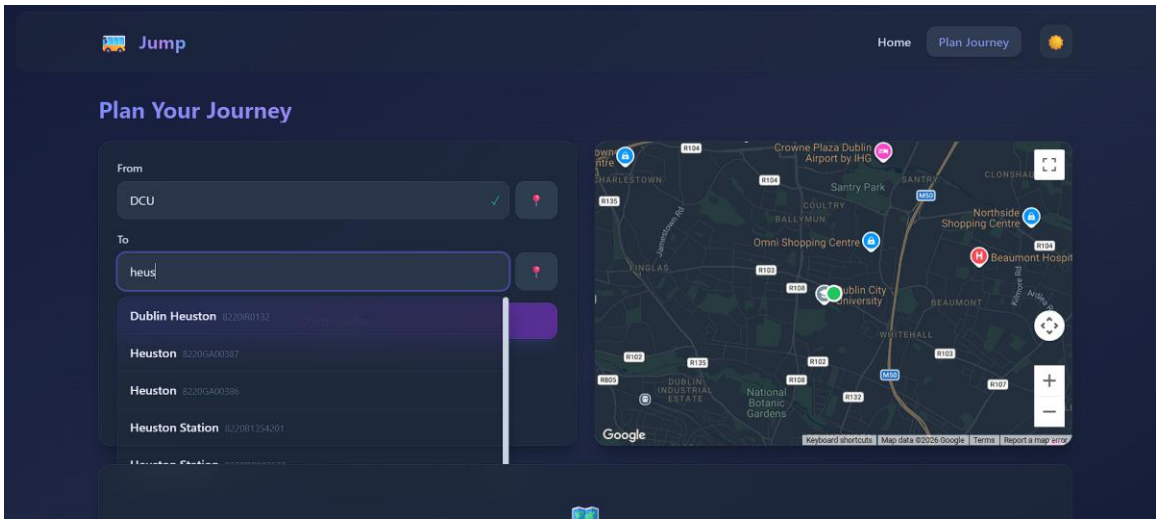
Click "Plan Journey" in the nav bar or the button on the home page. This is where you pick two locations and find routes between them.



Picking your start and end points

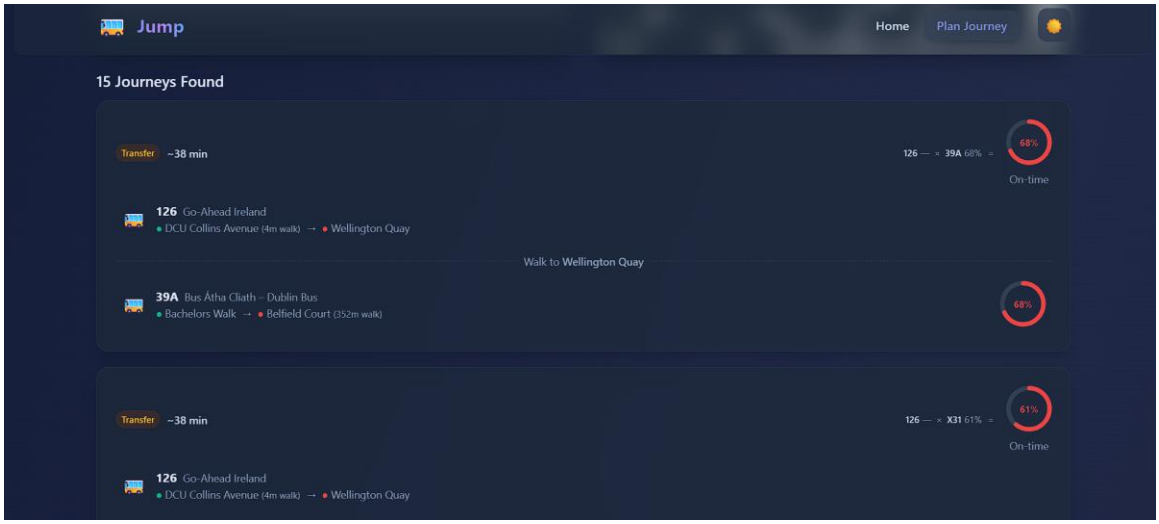
There are two text fields on the left: "From" and "To". You can either type a stop name and pick from the dropdown that appears, or click the little pin button next to the field and then click a spot on the map. Either way, when a location is set you'll see a green tick in the field.

For the dropdown, just start typing and suggestions pop up. You can use arrow keys and Enter to select, or just click one. Press Escape to close it.



Searching

Once both fields have green ticks, hit "Find Routes". It'll take a second to search.

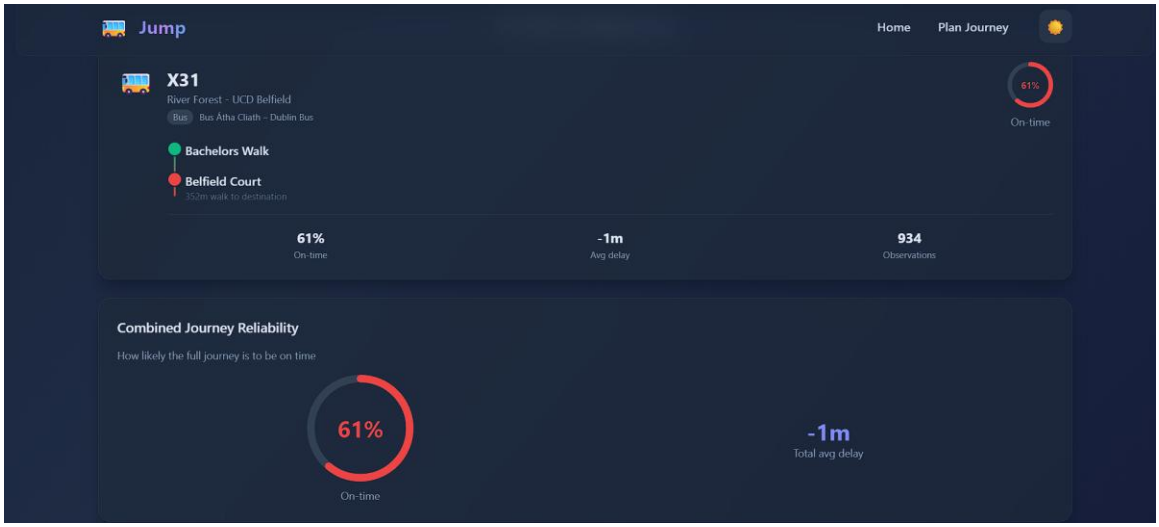


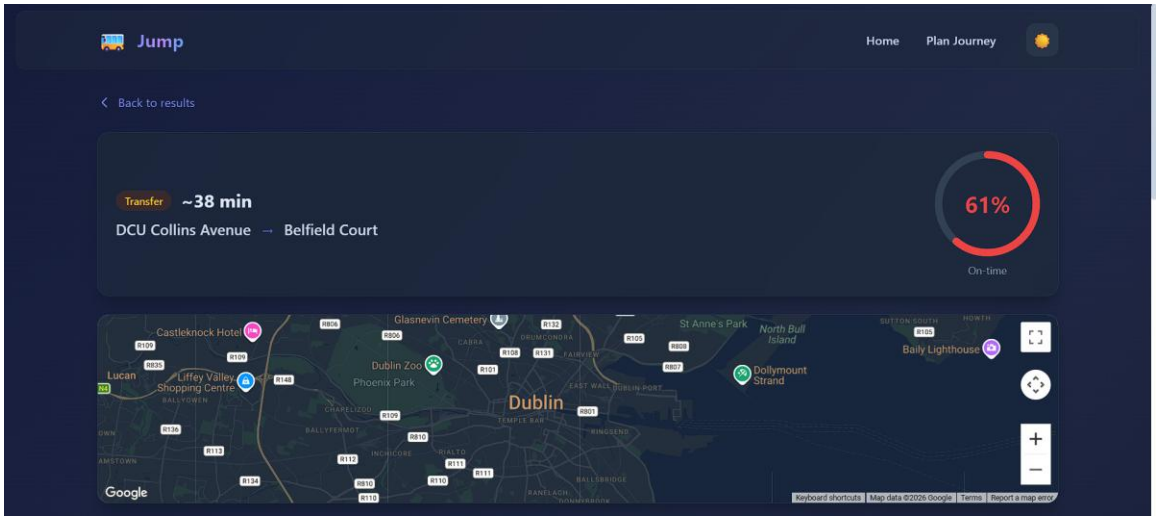
Results

Results come back as cards. Each one says "Direct" if it's a single route with no changes, or "Transfer" if you need to switch at some point.

On each card you can see the route number, operator, which stops you'd board and get off at, walking distances if you need to walk to/from a stop, and a reliability gauge. Transfer journeys also show the reliability of each leg separately so you can see which bit might let you down.

You can click any route name in the results to go to its full details page.




Jump
Home
Plan Journey
Back to results
Transfer
~38 min
DCU Collins Avenue → Belfield Court
61%
On-time
Castleknock Hotel
Lucan
Liffey Valley Shopping Centre
Glasnevin Cemetery
Dublin Zoo
Phoenix Park
Dublin
St Anne's Park
North Bull Island
Dollymount Strand
Baily Lighthouse
Google
Keyboard shortcuts
Map data ©2026 Google
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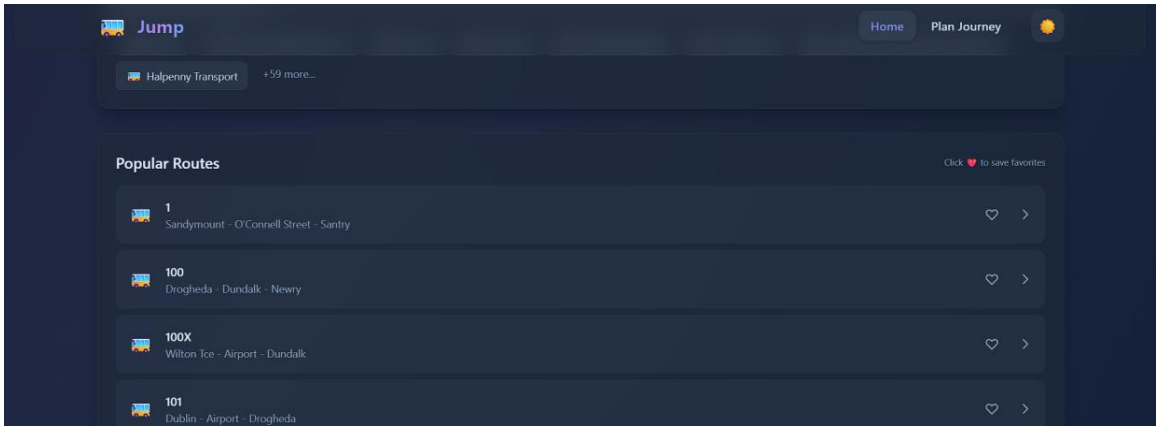
Report a map error

5. Route Details

Click on any route in the app and you'll land here. It's got everything about that one route.

At the top there's the route number, full name, a badge for the type (bus/rail/tram), and the operator. There's also a heart button to favourite it.



Reliability

This is the main bit. There's a big gauge showing the on-time percentage, plus the average delay and how many data points went into the calculation.

The two dropdown filters are really useful here. You can pick a day of the week and an hour of the day to narrow it down. So if you want to know how the 39A does on a Monday at 8am, you can check exactly that. Set them back to "All Days" / "All Hours" to see the overall picture.

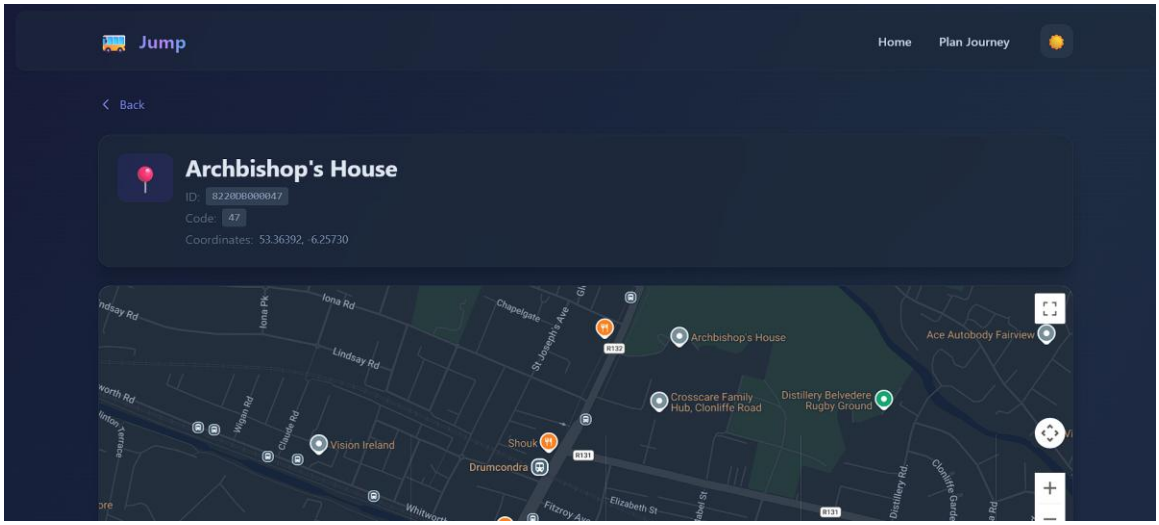
If there's no data for your chosen filter it'll say so, just try a different combo.

Map and Stops

There's a map with numbered markers for each stop on the route. Below that is a scrollable list of every stop in order. Click any stop to see its own details page.

6. Stop Details

This page is about one specific stop. You get here by clicking a stop name from a route details page.



It shows the stop name, ID, code, and coordinates. There's a map zoomed in on where the stop is, and a "View on Google Maps" link if you want to open it in Google Maps for directions.

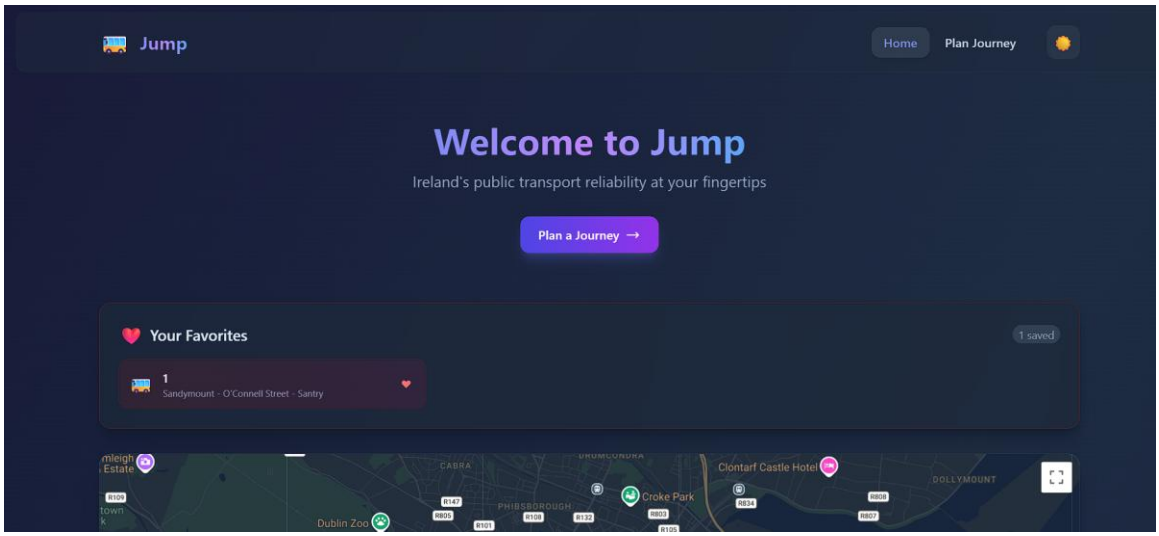
The best part is the "Routes at this stop" section. It lists every route that calls at this stop, each with its own reliability gauge. So if three different buses stop here, you can quickly compare which one is most reliable. Click any of them to go to the route details.

7. Favourites

If there's a route you check all the time, save it as a favourite. Just click the heart icon next to any route, anywhere in the app. It'll go red to show it's saved.

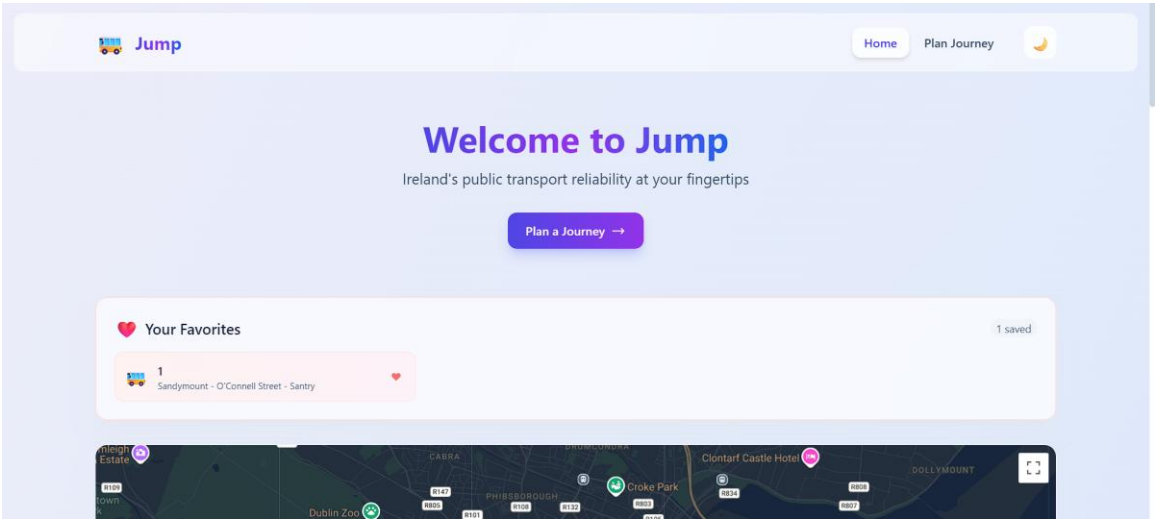
Your favourites show up in their own section at the top of the home page so you can get to them fast. To remove one, just click the heart again.

One thing to note: favourites are stored in your browser. If you clear your browsing data or use a different browser they won't carry over.



8. Dark Mode

There's a toggle in the top right of the nav bar. Click the moon to go dark, click the sun to go back to light. It remembers your choice for next time.



9. How Reliability Scores Work

The reliability gauges you see around the app are based on real data. We poll the NTA's live feeds and records when buses and trains are running ahead, on time, or behind schedule. We then compare that against the planned timetable.

The on-time percentage is the proportion of services that arrived within a reasonable window of their scheduled time. Green means good, amber is okay, red is poor.

On route details you can use the day and hour filters to dig into patterns. A route might be grand on weekends but terrible during rush hour, for example.

If a route shows no reliability data it just means the pipeline hasn't collected enough observations for it yet. Give it some time.

10. Common Issues

"Connection Error" when opening the app

The backend isn't running. If you're the one managing it, start it up with "python -m src.api.run" from the code folder. Make sure Docker is running too since the database lives there.

Map won't load

Probably a Google Maps API key issue. There should be a .env file in the dashboard folder with VITE_GOOGLE_MAPS_API_KEY set. If it's missing or the key is wrong the map won't show. Restart the frontend after fixing it.

No reliability data for a route

The data pipeline needs to have been running for a while to build up enough data. If it's a less popular route it might take longer. Try different day/hour filters or check back later.

Journey search finds nothing

Try picking stops that are closer to major roads or bus corridors. The search uses a radius around your selected points, so if they're out in the middle of nowhere there might not be stops nearby.

My favourites are gone

They're stored in browser local storage. If you cleared your browser data, or you're on a different browser or device, they won't be there. Unfortunately there's no way to get them back once cleared.